

Reinforcement Learning for Humanoid Locomotion

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Abstract—This report explores the application of reinforcement learning (RL) to achieve distinctive bipedal gaits in humanoid robots, while ensuring robustness and adaptability across varied terrains. It focuses on three methods: ‘Sim-to-Real Learning of All Common Bipedal Gaits,’ ‘Adapting Rapid Motor Adaptation for Bipedal Robots,’ and ‘Learning Humanoid Locomotion with Transformers.’ These strategies employ Proximal Policy Optimization (PPO) in simulations without visual sensor inputs, using dynamics randomization to address real-world disturbances. The goal is to elucidate these approaches, their execution, and their effects on bipedal robot mobility, thereby contributing to the ongoing discourse within the robotics field.

I. INTRODUCTION

The development of bipedal robots, aimed at achieving agility and precision in varied terrains similar to humans, has progressed since the first full-sized humanoid robot was introduced in the 1970s [1]. Progress in control systems [2]–[5] and optimization-based methods [6]–[9] has enhanced robots’ dynamic behaviors. Since the 1990s, Reinforcement Learning (RL) has enabled more advanced locomotion capabilities, improving terrain navigation for quadrupeds [10]–[13] and bipedal robots.

This report examines three RL-based strategies that have advanced bipedal locomotion: “Sim-to-Real Learning of All Common Bipedal Gaits” [14], [15], which achieves versatile gaits; “Adapting Rapid Motor Adaptation for Bipedal Robots” [16], focusing on quick adaptability to different terrains; and “Learning Humanoid Locomotion with Transformers” [17], [18], which uses an end-to-end learning approach. These methods reflect the evolution towards more adaptable and intelligent systems capable of addressing real-world challenges. The review provides an overview of these advancements, emphasizing their contributions to bipedal locomotion and future robotics implications.

II. METHODS

This section explores a range of innovative reinforcement learning (RL) methodologies designed for enhancing humanoid locomotion. It focuses on key developments in the field, as outlined below.

A. Sim-to-Real Learning of All Common Bipedal Gaits

1) *Approach Overview:* Traditional methods for learning bipedal gaits through reinforcement learning (RL) frequently face substantial obstacles. Methods based on reference trajectories [19]–[23] can replicate specific gait characteristics with a degree of accuracy but lack flexibility, which limits their ability to develop resilient and efficient gaits. Alternatively, reference-free strategies, such as those utilized in OpenAI

Gym [6], often result in an exhaustive and inefficient tuning process, failing to consistently achieve the desired outcomes.

Addressing these challenges, this study introduces a sim-to-real RL reward specification framework [14] designed for the autonomous acquisition of all prevalent bipedal gaits. This framework is distinguished by its independence from reference motions, instead utilizing a combination of straightforward, periodic cost functions based on forces and velocities to define gait-specific rewards. This novel method simplifies the adjustment of parameters for various gaits—including standing, walking, hopping, running, and skipping—thereby accurately aligning the reward function with the intended gait behavior.

A significant innovation of this framework is its employment of periodic reward functions that mirror the cyclical nature of bipedal movement. By focusing on the dynamics specific to swing and stance phases and introducing a probabilistic approach to apply cost functions at precise intervals, the framework promotes effective gait learning. This strategy eliminates the need for external benchmarks or constraints, marking a considerable advancement in the field of robotics by facilitating a more profound and naturalistic implementation of bipedal locomotion, as depicted in Figure 1.

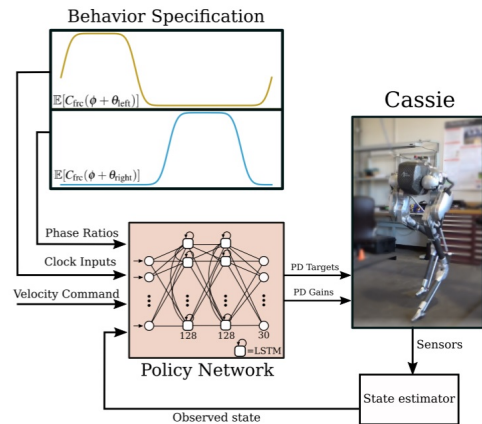


Fig. 1. Illustration of the reward design framework enabling the learning of locomotion behaviors such as standing, walking, and running in robots. This framework incorporates gait parameters into an LSTM policy to generate PD joint position targets and gains, optimizing robotic movements.

2) *Problem Formulation:* This research focuses on the sim-to-real transfer within the reinforcement learning (RL) domain, targeting bipedal gait policies. It defines a discrete-time Markov Decision Process (MDP) characterized by continuous state and action spaces (S and A), a transition function $T(s, a, s')$, and a non-stationary reward function

$R(s, t)$. The objective is to craft a control policy $\pi(a | s)$ aiming for the maximization of the expected T-horizon discounted return, $J(\pi)$, defined as:

$$J(\pi) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t R(S_t, t) \right] \quad (1)$$

where γ is the discount factor, and S_t represents the state at time t under policy π and transition dynamics T .

3) *Learning bipedal gaits(Periodic Reward Composition)*: Central to our approach is the periodic reward function, $R(s, \phi)$, dependent on a cycle time variable Φ , cycling from 0 to 1 in discrete time steps. This configuration enables the mapping of the reward function to Φ , applying costs cyclically:

$$R(s, \phi) = \beta + \sum_i R_i(s, \phi) \quad (2)$$

For each term $R_i(s, \phi)$, it incorporates a phase coefficient c_i , a phase indicator $I_i(\phi)$, and a measurement of phase-specific reward $q_i(s)$. his structure allows for targeted reward adjustments based on the gait phase, focusing on minimizing foot forces during swing phases to promote foot lifting, and reducing foot velocities during stance phases to ensure stability.

4) *Describing Bipedal Gaits*: The framework defines bipedal gaits by adjusting rewards for swing and stance phases, essential for bipedal movement. These phases involve either alternate or simultaneous actions of the left and right feet. Walking, for example, alternates the swing and stance phases between legs, ensuring continuous movement, while hopping involves both feet moving in sync, as shown in 2.

The formula for calculating the total expected reward from bipedal actions integrates the phase-specific rewards for each foot, considering the phase offsets (θ_{left} and θ_{right}) for the left and right feet:

$$\begin{aligned} \mathbb{E}[R_{\text{bipedal}}(s, \phi)] = & \mathbb{E}[C_{\text{frc}}(\phi + \theta_{\text{left}})] \cdot q_{\text{left frc}}(s) \\ & + \mathbb{E}[C_{\text{frc}}(\phi + \theta_{\text{right}})] \cdot q_{\text{right frc}}(s) \\ & + \mathbb{E}[C_{\text{spd}}(\phi + \theta_{\text{left}})] \cdot q_{\text{left spd}}(s) \\ & + \mathbb{E}[C_{\text{spd}}(\phi + \theta_{\text{right}})] \cdot q_{\text{right spd}}(s) \end{aligned} \quad (3)$$

5) *Model Architecture*: This study employs a Long Short-Term Memory (LSTM) neural network with two 128-unit recurrent hidden layers. The LSTM is selected for its ability to handle sequences, a crucial aspect for modeling bipedal locomotion which involves tracking the robot's state, command inputs, and time-dependent elements such as gait phase.

Network and Action Space: The output layer of the LSTM predicts 10 joint positions along with their Proportional-Derivative (PD) gains, targeting accurate control of the robot's movements. The network outputs at a rate of 40Hz, enabling timely interaction with the robot's PD controllers

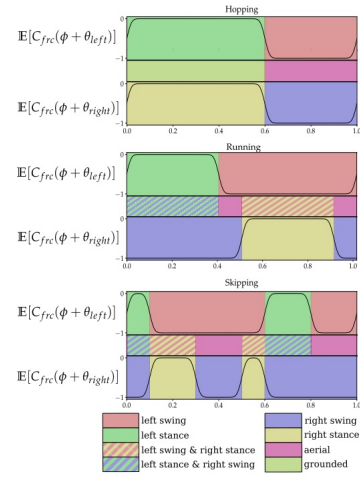


Fig. 2. Expected force phase coefficient for various bipedal gaits. Hopping displays negligible phase shift ($|\theta_{\text{left}} - \theta_{\text{right}}| \approx 0$) between feet. Walking and running show a moderate phase shift ($|\theta_{\text{left}} - \theta_{\text{right}}| \approx 0.5$), indicative of a galloping transition. Skipping requires four phases and mirrors the moderate shift ($|\theta_{\text{left}} - \theta_{\text{right}}| \approx 0.5$). Shaded areas represent hybrid phases combining behaviors of both feet.

operating at 2000Hz, for swift posture and dynamic adjustments.

State Space and Encoding: The state space includes the cycle time ϕ , the left and right cycle offset parameters (θ_{left} and θ_{right}), and timing information for each gait phase. It incorporates sine wave clocks for phase synchronization and a ratio vector for phase durations. These features enable the LSTM to efficiently handle temporal patterns and phase transitions, ensuring coordinated leg movements for bipedal locomotion.

6) *Policy Learning and Optimization*: Policies are optimized using Proximal Policy Optimization (PPO), tailored for bipedal locomotion. Training parameters include a batch size of 32 trajectories, each up to 300 timesteps, with a learning rate of 0.0001 applied to both actor and critic networks. A replay buffer stores 50,000 samples, undergoing training over 4 epochs per iteration, culminating after processing 150,000,000 samples. Training typically requires 24 to 36 hours per policy, employing the cassie-mujoco-sim simulator, which is based on MuJoCo, for simulating the Cassie robot's learning phase.

7) *Reward Function Design*: The reward function is concisely encapsulated as:

$$\mathbb{E}[R(s, \phi)] = \mathbb{E}[R_{\text{bipedal}}(s, \phi)] + R_{\text{smooth}}(s) + R_{\text{cmd}}(s) + \beta \quad (4)$$

where

$$R_{\text{cmd}}(s) = (-1) \cdot q_{i_x}(s) \quad (5)$$

$$+ (-1) \cdot q_{i_y}(s) \quad (6)$$

$$+ (-1) \cdot q_{\text{orientation}}(s) \quad (7)$$

$$R_{\text{smooth}}(s) = (-1) \cdot q_{\text{action diff}}(s) \quad (8)$$

$$+ (-1) \cdot q_{\text{torque}}(s) \quad (9)$$

$$+ (-1) \cdot q_{\text{pelvis acc}}(s) \quad (10)$$

$\mathbb{E}[R_{\text{bipedal}}(s, \phi)]$ quantifies the expected reward for bipedal gait patterns, emphasizing the cyclic aspects of movement.

$R_{\text{smooth}}(s)$ imposes penalties on sudden movements and high joint torques to encourage smoother, energy-efficient motions.

$R_{\text{cmd}}(s)$ rewards the robot's compliance with specified velocities and orientations, ensuring precise execution of commands.

Beta adjusts the reward scale to balance the encouragement of preferred gait characteristics with the deterrence of undesirable actions.

8) *Stair Climbing Extension:* To extend the reinforcement learning (RL) framework, initially focused on flat terrain, to include stair climbing capabilities, this study [15] details the necessary modifications for stair navigation. A key adaptation is the introduction of terrain randomization during training, which includes various stair configurations by altering height, width, and slope. This modification enhances the robot's proprioceptive control, allowing for autonomous navigation of stairs in both ascending and descending modes, as shown in 3.

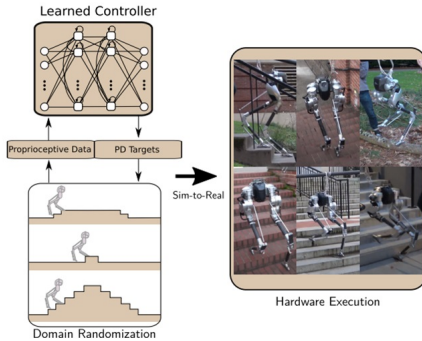


Fig. 3. Investigating the capabilities of blind bipedal locomotion, this research outlines a method for training policies that enable stair navigation without visual cues. These policies are designed to improve proprioceptive responses to significant ground height changes, facilitating reliable performance in a variety of settings.

B. Adapting Rapid Motor Adaptation for Bipedal Robots

1) *Approach Overview:* The Adapted Rapid Motor Adaptation (A-RMA) algorithm [16] is an extension of the RMA framework [25], originally developed for quadruped robots, to support bipedal robots. It initializes the base policy with reference motions from existing gait libraries [26], [27], overcoming the difficulty of starting from zero. To address

the heightened sensitivity of bipedal robots to environmental variances, A-RMA refines the policy using Proximal Policy Optimization (PPO), while maintaining a fixed adaptation module [28]. This method efficiently integrates initial policy bootstrapping with ongoing refinement, enhancing the adaptability of bipedal locomotion to changing environments, as shown in 4.

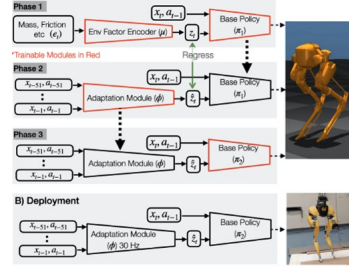


Fig. 4. Illustration of A-RMA's training and deployment phases, following RMA's framework [25] with an additional phase for fine-tuning the base policy using PPO, maintaining the adaptation module unchanged. This step precisely addresses external estimation inaccuracies.

2) *Bootstrapping the Base Policy:* The base policy for bipedal robots is initialized using reference motions from a Hybrid Zero Dynamics (HZD)-based gait library, facilitating the start of the learning process with a set of periodic walking gaits at various velocities and elevations. The policy π , given the current state x_t and an external vector $z \in \mathbb{R}^8$, estimates the next action a_t . This action specifies the target positions for the robot's actuated joints, which are then converted into torque by a PD controller. The vector z_t represents a streamlined version of a larger environmental vector e_t , reduced by the function μ to include only critical adaptation data:

$$z_t = \mu(e_t),$$

$$a_t = \pi_1(x_t, a_{t-1}, z_t).$$

This process is enhanced by implementing μ and π_1 as Multi-Layer Perceptrons (MLPs), jointly trained through model-free reinforcement learning to optimize the policy's expected return:

$$J(\pi) = \mathbb{E}_{\tau \sim p(\tau|\pi)} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right],$$

where τ represents the agent's trajectory under policy π_1 .

3) *Fine-tuning with PPO and Fixed Adaptation Module:* After the adaptation module's integration, it supports the base policy's adjustment using the external estimates. Recognizing the possible inaccuracies in these estimates by the adaptation module ϕ , a fine-tuning stage becomes necessary. This stage utilizes Proximal Policy Optimization (PPO) to enhance the base policy π_1 against the inaccurately estimated external factors, keeping the adaptation module ϕ unchanged. This fine-tuning is crucial to improve the base policy's handling of variances from imprecise external estimates, ensuring that the evolved base policy, π_2 , is

effectively adaptable to real-world conditions.

4) *Reward Function Design*: The reward function aims at optimizing motor position precision, pelvis movement imitation, torque efficiency, and ground reaction forces. It includes components to:

- 1) Motor imitation: $\exp(-\rho_1 \|q_m^r - \hat{q}_m\|_2^2)$
- 2) Pelvis position imitation: $\exp(-\rho_2 \|q_p - \hat{q}_p\|_2^2)$
- 3) Pelvis velocity imitation: $\exp(-\rho_3 \|\dot{q}_p^r - \dot{\hat{q}}_p\|_2^2)$
- 4) Pelvis Rot-imitation: $\exp(-\rho_4 (1 - \cos(q_T^r - \hat{q}_T^r)))$
- 5) Rotational velocity imitation: $\exp(-\rho_5 \|\dot{q}_T^r - \dot{\hat{q}}_T\|_2^2)$
- 6) Torque penalty: $\exp(-\rho_6 \|u\|_2^2)$
- 7) Ground reaction force penalty: $\exp(-\rho_7 \|F\|_2^2)$

C. Learning Humanoid Locomotion with Transformers

1) *Approach Overview*: This approach uses Transformers to enhance dynamic humanoid locomotion, aiming for immediate applicability in real-world scenarios. Transformers have been applied successfully in fields like natural language processing [29]–[31] and computer vision [32], [33], and now in robotics for tasks such as visual pre-training [34], creating object-centric representations [35], integrating depth information [36], and cloning behaviors conditioned on language instructions [37], [38]. This method utilizes Transformer models for online reinforcement learning in humanoid locomotion, independent of offline datasets.

Through simulation-based training, the robot learns to adjust its movement strategies, including arm swings for balance and adapting its gait to different terrains. This approach processes sequences of observations and actions for decision-making in partially observable environments. The training starts with optimizing a state policy in a simulated environment with full observability. It then uses Kullback-Leibler (KL) divergence for policy distillation into an observation policy, enhancing the robot's ability to adapt to environmental changes, as demonstrated in 5.

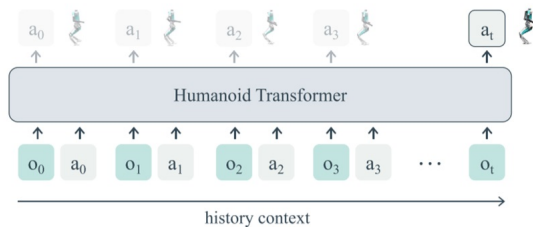


Fig. 5. Exploring blind bipedal locomotion limits, this study introduces a training pipeline for generating policies enabling real-world stair ascent and descent without visual input. These policies develop proprioceptive reflexes for substantial ground height disturbances, achieving robust performance across diverse environments.

2) *Problem Formulation*: The control challenge for humanoid locomotion is defined within a Markov Decision Process (MDP) framework and further expanded to a Partially Observable Markov Decision Process (POMDP) to address complexities in real-world scenarios.

MDP to POMDP Transition:

- **State Space(S)**: Describes potential states a robot might encounter, including various environmental configurations.
- **Action Space(A)**: Enumerates actions available to the robot, spanning a comprehensive range of movement options for navigation.
- **Transition Function $P(s_{t+1}|s_t, a_t)$** : Determines the likelihood of moving from one state to another after an action is taken, reflecting environmental variability.
- **Reward Function: $R(s_{t+1}|s_t, a_t)$** : Assigns a scalar value to each transition, assessing the action's impact.
- **Observation Space(O)**: Captures the robot's partial insights into the current state, considering sensor limitations and environmental factors.
- **Observation Function $Z(o_t|s_t)$** : Manages uncertainties in observations, crucial for operation in partially observable settings.

3) *Model Architecture: Transformer Model Implementation*: Transformers [39] facilitate the creation of control policies for humanoid robots in complex environments, modeled as Partially Observable Markov Decision Processes (POMDPs). These models process sequences of observation-action pairs, utilizing the sequential modeling capabilities of Transformers for enhanced action prediction. This adaptation of Transformers, traditionally employed in natural language processing and computer vision, to robotics emphasizes direct application to locomotion challenges in real-world settings without the need for offline datasets.

Self-Attention Mechanism: The self-attention mechanism, pivotal to Transformer models, allows for the weighted evaluation of input sequences, pinpointing dependencies vital for effective locomotion. It isolates significant features within observation-action pairs, aiding in the identification of dynamic movement strategies, including various gait patterns. Sinusoidal positional encodings are integrated to preserve the temporal context, thereby improving the model's ability to predict future actions with high temporal fidelity.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (11)$$

4) Policy Learning and Optimization:

State Policy Development: Developing a state policy in a fully observable simulated environment is the foundational step. It establishes a base policy (π_s) directing actions according to the present state, leveraging simulation for accelerated and effective training. Adjustments to reward

functions, including gait optimization, facilitate the derivation of an advanced policy for state-based decision-making, capturing essential locomotion and environmental interaction parameters.

Observation Policy Refinement: Refinement to an observation policy is achieved by applying Kullback-Leibler (KL) divergence, transitioning from a state-dependent to a partially observable decision-making framework. This process mirrors the teacher-student learning model [11], [12], adapting the policy for complexities and uncertainties encountered outside simulated conditions.

Joint Optimization Approach: A joint optimization strategy addresses the discrepancies between state and observation spaces, aligning decision-making processes. It merges reinforcement learning (RL) loss with state-policy supervision into a comprehensive objective function that harmonizes these aspects, ensuring an integrated approach to policy development.

$$L(\pi_o) = L_{RL}(\pi_o) + \lambda D_{KL}(\pi_o || \pi_s) \quad (12)$$

balances RL loss and the divergence between the observation and state policies, with the weighting factor.

5) Reward Function Design: The reward function for humanoid robot locomotion is structured to encompass a broad spectrum of dynamics and interactions within the environment. It focuses on velocity management by monitoring linear and angular velocities against set targets. Additionally, it evaluates stability and posture by rewarding base movement and orientation, aiming to maintain alignment with the gravitational vector. Interactions with the environment, including base height and foot contact quality, are optimized for safety and efficiency. The function prioritizes trajectory smoothness and joint motion efficiency, minimizing torques and accelerations for more natural movements. It also checks joint positional accuracy to ensure adherence to ideal configurations and integrates safety protocols to prevent collisions and operational errors.

To quantify these aspects, the function incorporates metrics like the linear and angular velocity tracking rewards, measuring compliance with desired velocities. The base motion reward penalizes vertical and roll-pitch deviations to enhance stability. The foot contact force reward evaluates ground interaction forces, ensuring they remain within safe and realistic limits. For comprehensive details on these and other components, refer to the specified literature.

$$r_v := \exp\left(-\frac{1}{2\sigma_{xy}} \|v_{xy} - v_{xy}^*\|^2\right) \quad (13)$$

$$r_{bm} := v_z^2 + 0.5 * \|\omega_{xy}\|^2 \quad (14)$$

$$r_f := \sum_{i \in \text{foot}} \begin{cases} \|F(i) - f_{\max}\|^2 & \text{if } \|F(i)\|^2 > f_{\max} \\ 0 & \text{if } \|F(i)\|^2 \leq f_{\max} \end{cases} \quad (15)$$

III. DISCUSSION

Feature	Sim-to-Real Gaits	Rapid Motor Adaptation	Humanoid Locomotion w/ Transformers
Core Concept	Dynamic gait system	Bootstrapping from gait library	Transformer-based sequential learning
Innovation	Periodic reward functions	Adaptation to environmental changes	Sequential decision-making
RL Algorithm	PPO	PPO	PPO
Architecture	LSTM networks	MLPs	Transformers
Adaptability	Moderate	High	High
Deployment Focus	Versatility in gaits	Rapid terrain adaptation	Zero-shot real-world application
Reward	Periodic costs for phases	Trajectory-based action replication	Comprehensive motion and stability tracking
Training	Sim. with dynamics randomization	PPO with static adaptation, model-free tuning	Large-scale RL, randomized sim. environments
Robot	Cassie	Cassie	Digit
Simulator	cassie-mujoco-sim	MuJoCo	Isaac Gym, AR-Sim

Sim-to-real Gaits Utilizing LSTM networks, this method enables humanoid robots to learn gait patterns without pre-defined trajectories, embracing the cyclic nature of bipedal locomotion. It successfully generates distinct behaviors such as hopping, walking, and running by fine-tuning ratios and cycle offsets, illustrating the method's adaptability.

Subsequent advancements introduced randomized stair scenarios in training, utilizing the same reward function. Analysis revealed that, compared to flat ground policies, the stair-trained policies enabled robots to learn adaptations such as increased foot clearance and a quicker leg angle retraction rate. However, these adaptations led to higher energy consumption, highlighting a trade-off between the adaptability of the two training strategies and their energy efficiency.

Outdoor experiments highlighted the real-world effectiveness of these policies, showcasing their robustness across varied terrains and conditions. The stair-trained LSTM policies particularly excelled in navigating stairs, demonstrating a high success rate in both ascending and descending trials, even under blind conditions. These policies managed dynamic tasks with impressive adaptability, such as hopping onto sidewalks, walking over curbs, and running over uneven surfaces without falling. The observed behaviors, including the ability to smoothly transition between gaits and adapt to the slope of the ground, underscore the policies' advanced force profile management over mere positional adjustments.

The outdoor tests confirmed the stair-trained policies' superior performance in real-world scenarios, especially in handling complex tasks like stair navigation, where they exhibited a higher overall probability of success compared to flat ground policies. This success rate, however, was influenced by the approach speed, indicating a nuanced balance between momentum and control in navigating stairs.

Rapid Motor Adaptation is based on leveraging a diverse gait library as a foundation for adaptive learning. This strategy enables immediate functionality and terrain adaptability through a structured yet flexible approach, providing initial stability while facilitating gradual environmental adaptation.

In real-world deployments on a Cassie bipedal robot, the A-RMA policy demonstrated remarkable adaptability across various challenging scenarios without the need for finetuning or calibration post-simulation training. The robot efficiently handled tasks like tracking variable commands on nominal ground and towing a heavy load, showcasing its capability to maintain stability and direction despite the added weight and variable pulling forces.

On slippery grounds covered with a plastic sheet and water, A-RMA successfully navigated significant slips and unexpected contact changes, maintaining gait stability and command tracking in both forward and lateral movements. This performance underscores the policy's ability to adapt to sudden changes in ground friction.

In tests involving rough terrains with variable softness, such as random placements of soft foams and wooden planks, A-RMA enabled Cassie to traverse these surfaces without losing balance. The robot adeptly adjusted to the differing contact types and softness levels, even though such conditions were not specifically included in its training simulations.

Humanoid Locomotion with Transformers leverages Transformer models for sequential decision-making, enhancing movement prediction accuracy by understanding temporal locomotion dynamics. This approach aims at improving adaptability in partially observable scenarios.

In real-world tests, the Transformer-based controller demonstrated significant stability across various scenarios, outperforming other neural network baselines in transferring learned behaviors to physical hardware. The robot successfully managed loads of different masses and shapes while maintaining stable gait and direction, highlighting the controller's capacity to adapt to added weight and balance challenges. This adaptability extended to handling sudden external forces, such as pushes or pulls, showcasing the controller's quick response to unexpected disturbances to maintain stability.

The controller's proficiency was further evident in navigating diverse terrains, from varying friction surfaces to rough terrains laden with potential obstructions, without exteroceptive sensing. It showed competence on slippery floors, and through obstacles like rubbers, cloths, and bubble wraps, underscoring its robustness and versatility in complex environments.

Most notably, the Transformer-based approach exhibited dynamic in-context adaptation, as seen when climbing steps not encountered during training, and responding to a simulated motor malfunction. The robot's ability to adjust its leg movements on subsequent attempts to climb steps, and its recovery from a motor malfunction without catastrophic failure, emphasizes the controller's reliance on historical

data for real-time adjustments.

Commonalities:

All three methods employ Proximal Policy Optimization (PPO) as their core learning algorithm, underlining PPO's effectiveness in balancing exploration and exploitation in complex environments. This choice is motivated by PPO's ability to efficiently handle the high-dimensional action spaces typical of humanoid robotics, ensuring stable learning progress.

To address sim-to-real discrepancies, each approach incorporates some form of Dynamics Randomization. This technique diversifies the simulation dynamics by randomizing physical parameters like damping, mass, and friction at the start of each training episode. Such randomization prevents overfitting to the simulated environment, facilitating a smoother transition to real-world applications. This strategy acknowledges the inevitable modeling errors in simulations and ensures robustness against them by exposing the learning algorithms to a wider range of dynamic conditions.

None of the methods rely on exteroceptive sensors, such as cameras, for terrain estimation. This design choice stems from the recognition that robots need to operate reliably in diverse and unpredictable real-world conditions, where reliance on such sensors could introduce vulnerability to occlusions, lighting variations, or other environmental factors. Instead, these approaches emphasize proprioception, aiming for a level of robustness that allows for effective navigation across a broad spectrum of human environments without detailed external sensory input.

Difference:

Adaptation Strategies to Diverse Terrains:

The Sim-to-Real Gaits method excels in pre-simulated, predictable environments by directly tuning to specific terrains, offering precise adaptability within known parameters. Its strength lies in customizing locomotion for anticipated scenarios, ensuring optimal performance in familiar settings but struggling with unexpected terrain changes.

Rapid Motor Adaptation (RMA) utilizes a gait library for flexibility, enabling swift adaptation to unpredictable terrains by leveraging a wide array of pre-learned movements. This approach combines the reliability of predefined gaits with on-the-fly learning's adaptability, making it suitable for environments with both known and novel terrain types.

Humanoid Locomotion with Transformers adopts an end-to-end learning strategy from historical data, dynamically adapting to new terrains. By analyzing complex patterns over time, this method provides nuanced adjustments to unforeseen environments, highlighting its capability for forward-looking adaptability without prior explicit programming for each potential scenario.

In summary, these methods illustrate different approaches to navigating the complex landscape of robotic locomotion. Sim-to-Real Gaits focuses on precision for known terrains, RMA balances pre-defined reliability with adaptive learning for mixed environments, and Transformers offer a forward-

thinking solution for unpredictable terrain changes. Each strategy reflects a unique balance between leveraging known gait patterns and adapting dynamically to the challenges of real-world mobility.

IV. CONCLUSIONS

This report synthesizes findings from key studies on "Reinforcement Learning for Humanoid Locomotion," focusing on three innovative strategies: Sim-to-Real Learning, Rapid Motor Adaptation, and Learning with Transformers. It outlines progress in increasing the adaptability and stability of bipedal robots, with each method addressing distinct aspects of navigation in real-world conditions. Sim-to-real learning ensures the effective application of simulation-trained behaviors, Rapid Motor Adaptation emphasizes quick adaptability to changing terrains, and Learning with Transformers offers sophisticated decision-making capabilities for dynamic locomotion. This analysis contributes to the scholarly discussion and sets a foundation for future investigations into optimizing humanoid robot movement.

V. FUTURE PERSPECTIVES

A. Integration of Multi-modal Sensory Data

Prospective research should consider the integration of multi-modal sensory data, such as visual and tactile feedback, into reinforcement learning algorithms for humanoid locomotion. This direction aims to enhance the environmental perception capabilities of robots, facilitating more sophisticated navigation and interaction within complex surroundings. The rationale behind this approach lies in its potential to closely mimic human sensory processing, thereby enabling robots to achieve higher levels of autonomy and environmental adaptability. Investigating this possibility is critical for advancing the field towards more versatile and intelligent humanoid robots.

B. Parallel Reinforcement Learning Across Multiple Robots

Future studies might also explore the implementation of parallel reinforcement learning processes across multiple humanoid robots, operating in varied environmental contexts. This innovative approach could lead to accelerated learning curves and the development of more robust locomotive strategies through the collective experience and data sharing among robots. The significance of this research lies in its potential to drastically improve the efficiency of learning algorithms and the generalization of learned behaviors, contributing to the scalability and applicability of humanoid robots in real-world scenarios.

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